

LAKSHIT TYAGI

Linux_Basics

lucky@ubuntu:~\$ cd Desktop/ (Used to Change Directory to Desktop)

lucky@ubuntu:~/Desktop\$ ls (View Contents of Desktop)

```
ai-research-summariser-infra  'Sample Travbrag'
'Azure Devops'                'Screenshot from 2025-09-03 23-47-35.png'
bbackuppp                    'Screenshot from 2025-09-13 03-40-49.png'
'Devops AWS.txt'              'Screenshot from 2025-09-13 03-42-39.png'
Docker                        testtt
Pipeline_For_ShellScript      TextFilesProject
Q1                            TravBrag
Q5                            'Travbrag Dump'
Recorded                      TravBragg-infra
Research-paper-summarizer-main. TravBrag.zip
Research-summarizer-k8s
```

lucky@ubuntu:~/Desktop\$ mkdir tute_dude (Creating a Dir for Tutedude Assignments)

lucky@ubuntu:~/Desktop\$ cd tute_dude/ (Opening dir)\

1. Creating and Renaming Files/Directories

1. Create a directory named test_dir using mkdir.

lucky@ubuntu:~/Desktop/tute_dude\$ mkdir test_dir (Making test_dir)

2. Inside test_dir, create an empty file called example.txt.

lucky@ubuntu:~/Desktop/tute_dude\$ ls test_dir (Listing Contents of DIR)

lucky@ubuntu:~/Desktop/tute_dude\$ cd test_dir/ (Opening DIR)

lucky@ubuntu:~/Desktop/tute_dude/test_dir\$ touch example.txt (Creating File)

lucky@ubuntu:~/Desktop/tute_dude/test_dir\$ ls (Listing Contents of DIR)
example.txt

3. Rename example.txt to renamed_example.txt using mv.

lucky@ubuntu:~/Desktop/tute_dude/test_dir\$ mv example.txt renamed_example.txt (Rename)

lucky@ubuntu:~/Desktop/tute_dude/test_dir\$ ls (Listing Contents of DIR)
renamed_example.txt

lucky@ubuntu:~/Desktop/tute_dude/test_dir\$

The screenshot shows a Google Docs document titled "Assignment_1_Linux_Basics.docx" and a terminal window. The document contains instructions for a Linux assignment, including a note that terminal commands and output are mandatory and should be shared as a repository link. The terminal window shows the user creating a directory structure and listing files.

(terminal / commands / output) are mandatory. Also share repository link.

Do NOT upload separate files.

Do NOT submit without compressing into a .zip file.

Objective: Practise essential Linux command-line operations: directory management, viewing files, searching, archiving, downloading, permissions, and environment variables.

1. Creating and Renaming Files/Directories

1. Create a directory named `test_dir` using `mkdir`.

2. Viewing File Contents

2. Inside `test_dir`, create an empty file called `example.txt`.
3. Rename `example.txt` to `renamed_example.txt` using `mv`.

3. Searching for Patterns

7. Use `grep` to find all lines containing the word "root" in `/etc/passwd`.

```
lucky@ubuntu:~$ cd Desktop/
lucky@ubuntu:~/Desktop$ ls
ai-research-summariser-infra  'Sample Travbrag'
'Azure Devops'                'Screenshot from 2025-09-03 23-47-35.png'
'bbackuppp                    'Screenshot from 2025-09-13 03-40-49.png'
'Devops AWS.txt'              'Screenshot from 2025-09-13 03-42-39.png'
Docker                        testttt
Pipeline_For_ShellScript      TextFilesProject
Q1                             Travbrag
Q5                             'Travbrag Dump'
Recorded                      Travbragg-infra
Research-paper-summarizer-main. Travbragg.zip
research-summarizer-k8s
lucky@ubuntu:~/Desktop$ mkdir tute_dude
lucky@ubuntu:~/Desktop$ cd tute_dude/
lucky@ubuntu:~/Desktop/tute_dude$ mkdir test_dir
lucky@ubuntu:~/Desktop/tute_dude$ ls
test_dir
lucky@ubuntu:~/Desktop/tute_dude$ cd test_dir/
lucky@ubuntu:~/Desktop/tute_dude/test_dir$ touch example.txt
lucky@ubuntu:~/Desktop/tute_dude/test_dir$ ls
example.txt
lucky@ubuntu:~/Desktop/tute_dude/test_dir$ mv example.txt renamed_example.txt
lucky@ubuntu:~/Desktop/tute_dude/test_dir$ ls
renamed_example.txt
lucky@ubuntu:~/Desktop/tute_dude/test_dir$
```

2. Viewing File Contents

1. Use `cat` to display the contents of `/etc/passwd`.

```
lucky@ubuntu:~/Desktop/tute_dude/test_dir$ cat /etc/passwd
```

```
root:x:0:0:root:/root:/bin/bash
```

```
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
```

```
bin:x:2:2:bin:/bin:/usr/sbin/nologin
```

```
sys:x:3:3:sys:/dev:/usr/sbin/nologin
```

```
sync:x:4:65534:sync:/bin:/bin/sync
```

```
games:x:5:60:games:/usr/games:/usr/sbin/nologin
```

```
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
```

```
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
```

```
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
```

```
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
```

uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534:/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:101:/nonexistent:/usr/sbin/nologin
syslog:x:102:102:/nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:/:/usr/sbin/nologin
uidd:x:103:103:/run/uidd:/usr/sbin/nologin
usbmux:x:104:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
tss:x:105:105:TPM software stack,,,:/var/lib/tpm:/bin/false
systemd-oom:x:990:990:systemd Userspace OOM Killer:/:/usr/sbin/nologin
kernoops:x:106:65534:Kernel Oops Tracking Daemon,,,:/usr/sbin/nologin
whoopsie:x:107:109:/nonexistent:/bin/false
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
avahi:x:108:111:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
tcpdump:x:109:112:/nonexistent:/usr/sbin/nologin
sssd:x:110:113:SSSD system user,,,:/var/lib/sss:/usr/sbin/nologin
speech-dispatcher:x:111:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false

cups-pk-helper:x:112:114:user for cups-pk-helper
service,,,:/nonexistent:/usr/sbin/nologin

fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin

saned:x:113:116:./var/lib/saned:/usr/sbin/nologin

geoclue:x:114:117:./var/lib/geoclue:/usr/sbin/nologin

cups-browsed:x:115:114:./nonexistent:/usr/sbin/nologin

hplip:x:116:7:HPLIP system user,,,:/run/hplip:/bin/false

gnome-remote-desktop:x:988:988:GNOME Remote
Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin

polkitd:x:987:987:User for polkitd:/usr/sbin/nologin

rtkit:x:117:119:RealtimeKit,,,:/proc:/usr/sbin/nologin

colord:x:118:120:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin

gnome-initial-setup:x:119:65534:./run/gnome-initial-setup:/bin/false

gdm:x:120:121:Gnome Display Manager:/var/lib/gdm3:/bin/false

nm-openvpn:x:121:122:NetworkManager
OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin

lukky:x:1000:1000:Lakshit Tyagi:/home/lukky:/bin/bash

sshd:x:124:65534:./run/sshd:/usr/sbin/nologin

slave1:x:1001:1001:./home/slave1:/bin/bash

arun:x:1002:1002:./home/arun:/bin/sh

ramesh:x:1003:1003,,,:/home/ramesh:/bin/bash

jenkins:x:123:125:Jenkins,,,:/var/lib/jenkins:/bin/bash

postfix:x:125:126:./var/spool/postfix:/usr/sbin/nologin

mongodb:x:126:65534:./nonexistent:/usr/sbin/nologin

nvidia-persistenced:x:122:124:NVIDIA Persistence
Daemon,,,:/nonexistent:/usr/sbin/nologin

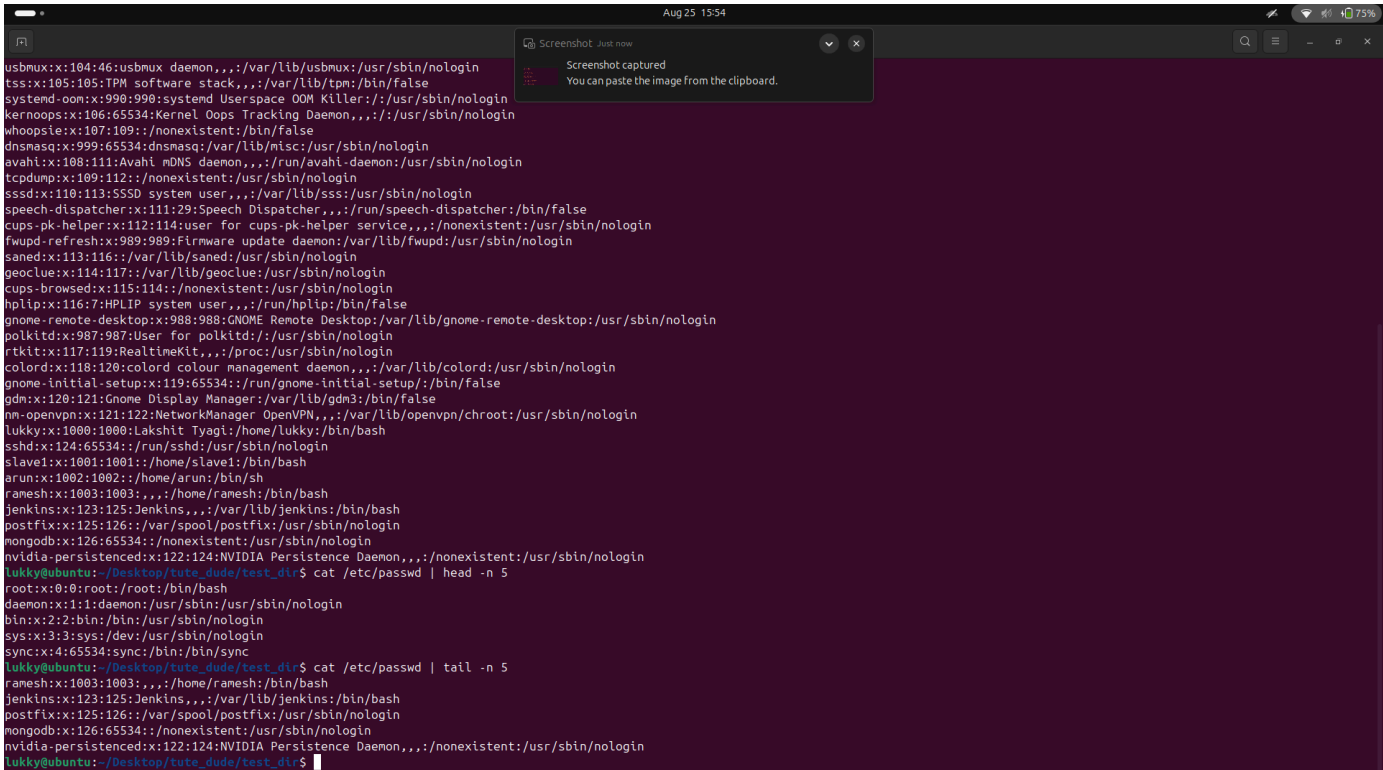
```
Aug 25 15:54
lucky@ubuntu: ~/Desktop/tute_dude/test_dir$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mail List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-networkd:x:998:998:systemd Network Management:/usr/sbin/nologin
systemd-timesyncd:x:996:996:systemd Time Synchronization:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd/bin/false
messagebus:x:101:101::/nonexistent:/usr/sbin/nologin
syslog:x:102:102::/nonexistent:/usr/sbin/nologin
systemd-resolved:x:991:991:systemd Resolver:/usr/sbin/nologin
uuidd:x:103:103::/run/uuidd:/usr/sbin/nologin
usbmux:x:104:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
tss:x:105:105:TPM software stack,,,:/var/lib/tpm:/bin/false
systemd-oom:x:990:990:systemd Userspace OOM Killer:/usr/sbin/nologin
kernoops:x:106:65534:Kernel Oops Tracking Daemon,,,:/usr/sbin/nologin
whoopsie:x:107:109::/nonexistent:/bin/false
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
avahi:x:108:111:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
tcpdump:x:109:112::/nonexistent:/usr/sbin/nologin
sssd:x:110:113:SSSD system user,,,:/var/lib/sss:/usr/sbin/nologin
speech-dispatcher:x:111:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
cups-pk-helper:x:112:114:user for cups-pk-helper service,,,:/nonexistent:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
saned:x:113:116:/var/lib/saned:/usr/sbin/nologin
geoclue:x:114:117:/var/lib/geoclue:/usr/sbin/nologin
cups-browsed:x:115:114::/nonexistent:/usr/sbin/nologin
hplip:x:116:7:HPLIP system user,,,:/run/hplip:/bin/false
gnome-remote-desktop:x:988:988:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin
polkitd:x:987:987>User for polkitd:/usr/sbin/nologin
rtkit:x:117:119:RealtimeKit,,,:/proc:/usr/sbin/nologin
```

2. Display only the first 5 lines of `/etc/passwd` using `head`.

```
lucky@ubuntu:~/Desktop/tute_dude/test_dir$ cat /etc/passwd | head -n 5
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
```

3. Display only the last 5 lines of `/etc/passwd` using `tail`.

```
lucky@ubuntu:~/Desktop/tute_dude/test_dir$ cat /etc/passwd | tail -n 5
ramesh:x:1003:1003::,/home/ramesh:/bin/bash
jenkins:x:123:125:Jenkins,,,:/var/lib/jenkins:/bin/bash
postfix:x:125:126:/var/spool/postfix:/usr/sbin/nologin
mongodb:x:126:65534::/nonexistent:/usr/sbin/nologin
nvidia-persistenced:x:122:124:NVIDIA Persistence Daemon,,,:/nonexistent:/usr/sbin/nologin
```



```
usbmux:x:104:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
tss:x:105:105:TPM software stack,,,:/var/lib/tpm:/bin/false
systemd-oom:x:990:990:systemd Userspace OOM Killer:/usr/sbin/nologin
kernoops:x:106:65534:Kernel Oops Tracking Daemon,,,:/usr/sbin/nologin
whoopsle:x:107:109::/nonexistent:/bin/false
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
avahi:x:108:111:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
tcpdump:x:109:112:/nonexistent:/usr/sbin/nologin
sssd:x:110:113:SSSD system user,,,:/var/lib/sss:/usr/sbin/nologin
speech-dispatcher:x:111:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
cups-pk-helper:x:112:114:user for cups-pk-helper service,,,:/nonexistent:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
saned:x:113:116::/var/lib/saned:/usr/sbin/nologin
geoclue:x:114:117::/var/lib/geoclue:/usr/sbin/nologin
cups-browsed:x:115:114:/nonexistent:/usr/sbin/nologin
hplip:x:116:7:HPLIP system user,,,:/run/hplip:/bin/false
gnome-remote-desktop:x:988:988:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/usr/sbin/nologin
rtkit:x:117:119:RealtimeKit,,,:/proc:/usr/sbin/nologin
colord:x:118:120:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
gnome-initial-setup:x:119:65534:/run/gnome-initial-setup:/bin/false
gdm:x:120:121:Gnome Display Manager:/var/lib/gdm3:/bin/false
nm-openvpn:x:121:122:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
lucky:x:1000:1000:Lakshit Tyagi:/home/lucky:/bin/bash
sshd:x:124:65534:/run/sshd:/usr/sbin/nologin
slave1:x:1001:1001:/home/slave1:/bin/bash
arun:x:1002:1002:/home/arun:/bin/sh
ramesh:x:1003:1003:::/home/ramesh:/bin/bash
jenkins:x:123:125:Jenkins,,,:/var/lib/jenkins:/bin/bash
postfix:x:125:126:/var/spool/postfix:/usr/sbin/nologin
mongod:x:126:65534:/nonexistent:/usr/sbin/nologin
nvidia-persistenced:x:122:124:NVIDIA Persistence Daemon,,,:/nonexistent:/usr/sbin/nologin
lucky@ubuntu:~/Desktop/tute_dude/test_dir$ cat /etc/passwd | head -n 5
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
lucky@ubuntu:~/Desktop/tute_dude/test_dir$ cat /etc/passwd | tail -n 5
ramesh:x:1003:1003:::/home/ramesh:/bin/bash
jenkins:x:123:125:Jenkins,,,:/var/lib/jenkins:/bin/bash
postfix:x:125:126:/var/spool/postfix:/usr/sbin/nologin
mongod:x:126:65534:/nonexistent:/usr/sbin/nologin
nvidia-persistenced:x:122:124:NVIDIA Persistence Daemon,,,:/nonexistent:/usr/sbin/nologin
lucky@ubuntu:~/Desktop/tute_dude/test_dir$
```

3. Searching for Patterns

1. Use `grep` to find all lines containing the word “root” in `/etc/passwd`.

```
lucky@ubuntu:~/Desktop/tute_dude/test_dir$ cat /etc/passwd | grep root
```

```
root:x:0:0:root:/root:/bin/bash
```

```
nm-openvpn:x:121:122:NetworkManager
OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
```

```
Aug 25 15:57
lucky@ubuntu: ~/Desktop/tute_dude/test_dir

whoopsie:x:107:109::/nonexistent:/bin/false
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
avahi:x:108:111:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
tcpdump:x:109:112::/nonexistent:/usr/sbin/nologin
sssd:x:110:113:SSSD system user,,,:/var/lib/sss:/usr/sbin/nologin
speech-dispatcher:x:111:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
cups-pk-helper:x:112:114:user for cups-pk-helper service,,,:/nonexistent:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
saned:x:113:116::/var/lib/saned:/usr/sbin/nologin
geoclue:x:114:117::/var/lib/geoclue:/usr/sbin/nologin
cups-browsed:x:115:114::/nonexistent:/usr/sbin/nologin
hplip:x:116:7:HPLIP system user,,,:/run/hplip:/bin/false
gnome-remote-desktop:x:988:988:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/usr/sbin/nologin
rtkit:x:117:119:RealtimeKit,,,:/proc:/usr/sbin/nologin
colord:x:118:120:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
gnome-initial-setup:x:119:65534::/run/gnome-initial-setup:/bin/false
gdm:x:120:121:Gnome Display Manager:/var/lib/gdm3:/bin/false
nm-openvpn:x:121:122:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
lucky:x:1000:1000:Lakshit Tyagi:/home/lucky:/bin/bash
sshd:x:124:65534::/run/sshd:/usr/sbin/nologin
slave1:x:1001:1001::/home/slave1:/bin/bash
arun:x:1002:1002::/home/arun:/bin/sh
ramesh:x:1003:1003::/home/ramesh:/bin/bash
jenkins:x:123:125:Jenkins,,,:/var/lib/jenkins:/bin/bash
postfix:x:125:126:/var/spool/postfix:/usr/sbin/nologin
mongodb:x:126:65534::/nonexistent:/usr/sbin/nologin
nvidia-persistenced:x:122:124:NVIDIA Persistence Daemon,,,:/nonexistent:/usr/sbin/nologin
lucky@ubuntu:~/Desktop/tute_dude/test_dir$ cat /etc/passwd | head -n 5
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
lucky@ubuntu:~/Desktop/tute_dude/test_dir$ cat /etc/passwd | tail -n 5
ramesh:x:1003:1003::/home/ramesh:/bin/bash
jenkins:x:123:125:Jenkins,,,:/var/lib/jenkins:/bin/bash
postfix:x:125:126:/var/spool/postfix:/usr/sbin/nologin
mongodb:x:126:65534::/nonexistent:/usr/sbin/nologin
nvidia-persistenced:x:122:124:NVIDIA Persistence Daemon,,,:/nonexistent:/usr/sbin/nologin
lucky@ubuntu:~/Desktop/tute_dude/test_dir$ cat /etc/passwd | grep root
root:x:0:0:root:/root:/bin/bash
nm-openvpn:x:121:122:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
lucky@ubuntu:~/Desktop/tute_dude/test_dir$ ^C
lucky@ubuntu:~/Desktop/tute_dude/test_dir$
```

4. Zipping and Unzipping

1. Compress the test_dir directory into a file named test_dir.zip using zip.
lucky@ubuntu:~/Desktop/tute_dude\$ sudo apt install zip unzip (**Installing zip/unzip**)

Reading package lists... Done

Building dependency tree... Done

Reading state information... Done

zip is already the newest version (3.0-13ubuntu0.2).

unzip is already the newest version (6.0-28ubuntu4.1).

The following packages were automatically installed and are no longer required:

libfwupd2 libslirp0 libwoff1 linux-headers-6.17.0-19-generic
linux-hwe-6.17-headers-6.17.0-19 linux-hwe-6.17-tools-6.17.0-19
linux-image-6.17.0-19-generic linux-modules-6.17.0-19-generic

linux-modules-extra-6.17.0-19-generic linux-objects-nvidia-580-6.17.0-19-generic
linux-signatures-nvidia-6.17.0-19-generic linux-tools-6.17.0-19-generic slirp4netns

Use 'sudo apt autoremove' to remove them.

0 upgraded, 0 newly installed, 0 to remove and 16 not upgraded.

lucky@ubuntu:~/Desktop/tute_dude\$ zip test_dir.zip test_dir **(Zipping the file)**

adding: test_dir/ (stored 0%)

lucky@ubuntu:~/Desktop/tute_dude\$ ls **(Listing Contents of DIR)**

test_dir test_dir.zip

2. Unzip test_dir.zip into a new directory named unzipped_dir.

lucky@ubuntu:~/Desktop/tute_dude\$ mkdir unzipped_dir **(Creating DIR)**

lucky@ubuntu:~/Desktop/tute_dude\$ unzip test_dir.zip -d unzipped_dir/

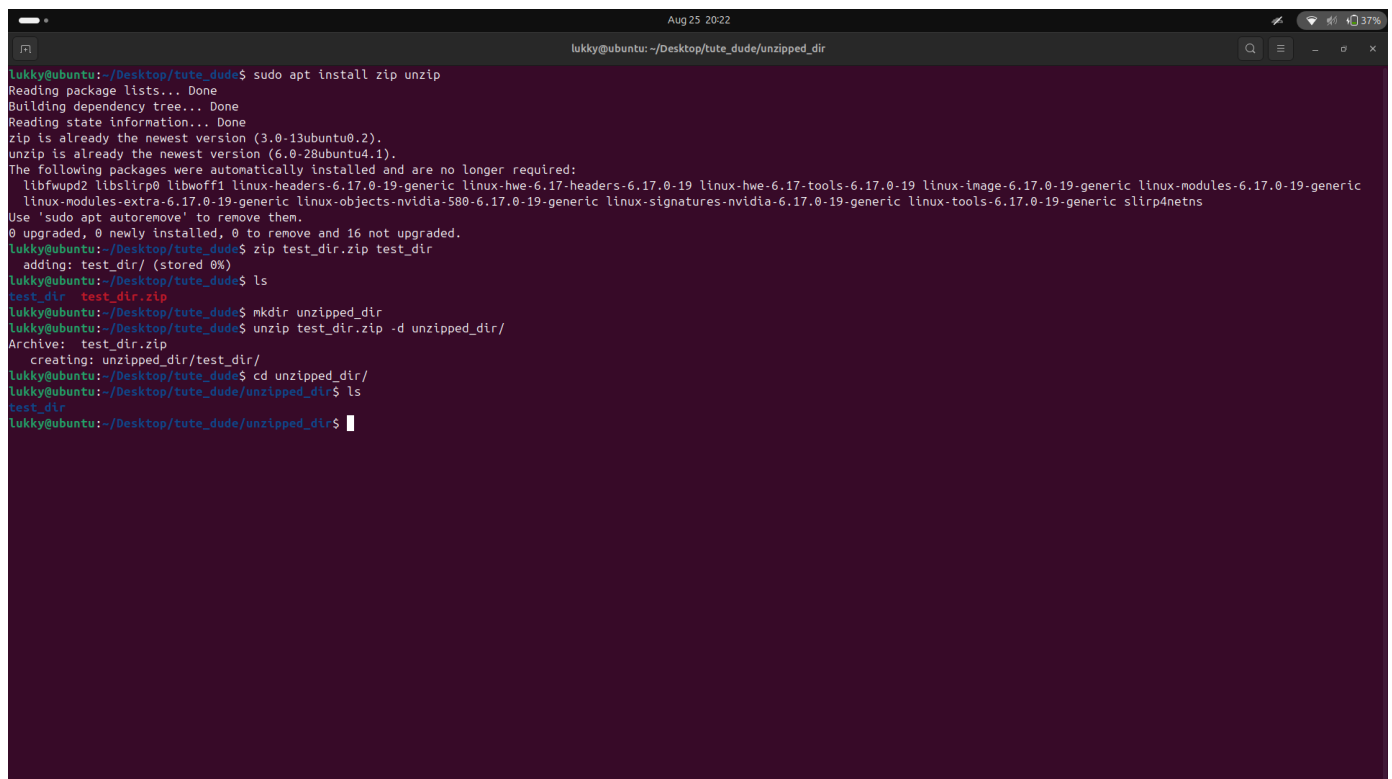
Archive: test_dir.zip

creating: unzipped_dir/test_dir/

lucky@ubuntu:~/Desktop/tute_dude\$ cd unzipped_dir/

lucky@ubuntu:~/Desktop/tute_dude/unzipped_dir\$ ls

test_dir



```
Aug 25 20:22
lucky@ubuntu:~/Desktop/tute_dude/unzipped_dir

lucky@ubuntu:~/Desktop/tute_dude$ sudo apt install zip unzip
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
zip is already the newest version (3.0-13ubuntu0.2).
unzip is already the newest version (6.0-28ubuntu4.1).
The following packages were automatically installed and are no longer required:
  libfwupd2 libslirp0 libwoff1 linux-headers-6.17.0-19-generic linux-hwe-6.17-headers-6.17.0-19 linux-hwe-6.17-tools-6.17.0-19 linux-image-6.17.0-19-generic linux-modules-6.17.0-19-generic
  linux-modules-extra-6.17.0-19-generic linux-objects-nvidia-580-6.17.0-19-generic linux-signatures-nvidia-6.17.0-19-generic linux-tools-6.17.0-19-generic slirp4netns
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 16 not upgraded.
lucky@ubuntu:~/Desktop/tute_dude$ zip test_dir.zip test_dir
  adding: test_dir/ (stored 0%)
lucky@ubuntu:~/Desktop/tute_dude$ ls
test_dir  test_dir.zip
lucky@ubuntu:~/Desktop/tute_dude$ mkdir unzipped_dir
lucky@ubuntu:~/Desktop/tute_dude$ unzip test_dir.zip -d unzipped_dir/
Archive: test_dir.zip
  creating: unzipped_dir/test_dir/
lucky@ubuntu:~/Desktop/tute_dude$ cd unzipped_dir/
lucky@ubuntu:~/Desktop/tute_dude/unzipped_dir$ ls
test_dir
lucky@ubuntu:~/Desktop/tute_dude/unzipped_dir$
```

5. Downloading Files

1. Use `wget` to download a file from a URL (e.g., `https://example.com/sample.txt`).

```
lucky@ubuntu:~/Desktop/tute_dude$ wget
https://raw.githubusercontent.com/Lukky175/ai-research-summariser-infra/main/readme.md
```

```
--2026-08-25 20:30:58--
https://raw.githubusercontent.com/Lukky175/ai-research-summariser-infra/main/readme.md
```

```
Resolving raw.githubusercontent.com (raw.githubusercontent.com)...
2606:50c0:8000::154, 2606:50c0:8001::154, 2606:50c0:8002::154, ...
```

```
Connecting to raw.githubusercontent.com
(raw.githubusercontent.com)[2606:50c0:8000::154]:443... connected.
```

```
HTTP request sent, awaiting response... 200 OK
```

```
Length: 1458 (1.4K) [text/plain]
```

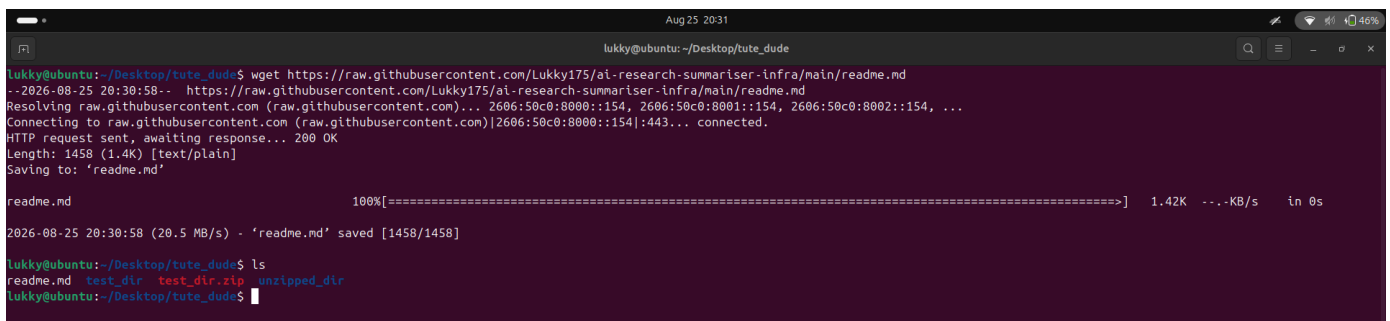
```
Saving to: 'readme.md'
```

```
readme.md
100%[=====] 1.42K --.-KB/s in 0s
=====>]
```

```
2026-08-25 20:30:58 (20.5 MB/s) - 'readme.md' saved [1458/1458]
```

```
lucky@ubuntu:~/Desktop/tute_dude$ ls
```

```
readme.md test_dir test_dir.zip unzipped_dir
```



```
Aug 25 20:31
lucky@ubuntu: ~/Desktop/tute_dude

lucky@ubuntu:~/Desktop/tute_dude$ wget https://raw.githubusercontent.com/Lukky175/ai-research-summariser-infra/main/readme.md
--2026-08-25 20:30:58-- https://raw.githubusercontent.com/Lukky175/ai-research-summariser-infra/main/readme.md
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 2606:50c0:8000::154, 2606:50c0:8001::154, 2606:50c0:8002::154, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)[2606:50c0:8000::154]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1458 (1.4K) [text/plain]
Saving to: 'readme.md'

readme.md
100%[=====] 1.42K --.-KB/s in 0s

2026-08-25 20:30:58 (20.5 MB/s) - 'readme.md' saved [1458/1458]

lucky@ubuntu:~/Desktop/tute_dude$ ls
readme.md test_dir test_dir.zip unzipped_dir
lucky@ubuntu:~/Desktop/tute_dude$
```

6. Changing Permissions

1. Create a file named `secure.txt` and change its permissions to read-only for everyone using `chmod`.

```
lucky@ubuntu:~/Desktop/tute_dude$ touch secure.txt (Creating File)
```

```
lucky@ubuntu:~/Desktop/tute_dude$ chmod 444 secure.txt (Changing Perms)
```

```
lucky@ubuntu:~/Desktop/tute_dude$ ls -l (Check Permissions)
```

```
total 16
```

```
-rw-rw-r-- 1 lucky lucky 1458 Aug 25 20:30 readme.md
```

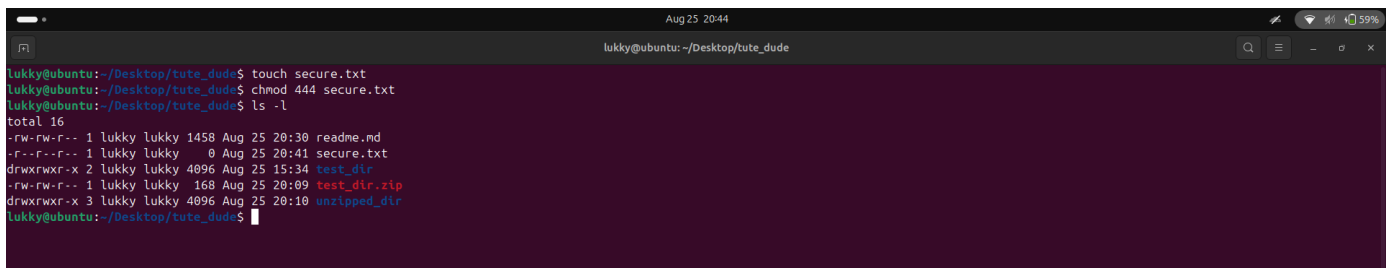
```
-r--r--r-- 1 lucky lucky 0 Aug 25 20:41 secure.txt
```

```
drwxrwxr-x 2 lucky lucky 4096 Aug 25 15:34 test_dir
```

```
-rw-rw-r-- 1 lucky lucky 168 Aug 25 20:09 test_dir.zip
```

```
drwxrwxr-x 3 lucky lucky 4096 Aug 25 20:10 unzipped_dir
```

```
lucky@ubuntu:~/Desktop/tute_dude$
```

A terminal window screenshot showing the execution of three commands: `touch secure.txt`, `chmod 444 secure.txt`, and `ls -l`. The output of `ls -l` shows the file `secure.txt` with permissions `-r--r--r--`. The terminal window has a title bar with the date and time "Aug 25 20:44" and a search icon.

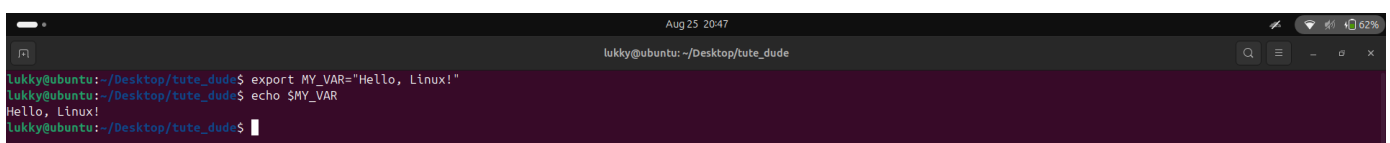
7. Working with Environment Variables

1. Use `export` to set a new environment variable called `MY_VAR` with the value "Hello, Linux!".

```
lucky@ubuntu:~/Desktop/tute_dude$ export MY_VAR="Hello, Linux!"
```

```
lucky@ubuntu:~/Desktop/tute_dude$ echo $MY_VAR
```

```
Hello, Linux!
```

A terminal window screenshot showing the execution of two commands: `export MY_VAR="Hello, Linux!"` and `echo $MY_VAR`. The output of `echo $MY_VAR` is "Hello, Linux!". The terminal window has a title bar with the date and time "Aug 25 20:47" and a search icon.